

The Floating Neutral: Endogenous Reference Theory and Moral Reference Without Authority

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Abstract

Metaethical realism and anti-realism share an unexamined assumption: that a moral *reference* — the standard against which actions are judged — requires an *anchor*. The realist seeks the anchor in mind-independent binding facts (or in God); the anti-realist, finding no such anchor, concludes there is no determinate moral fact at all, only projection, error, or arbitrary commitment. We reject the shared assumption. Drawing on the behavior of a **floating neutral** in a polyphase electrical system, we develop **Endogenous Reference Theory (ERT)**: the moral reference of a system is a determinate, computable point fixed entirely by the system’s own constituents — its agents and their weighted valuations — with no external anchor and *no normative authority*. The reference has a *location* but issues no *commands*. On this view “ought” does not name a binding force transmitted from a higher level; it names **displacement** from the endogenous reference — a strain, not a decree. ERT makes three contributions the existing literature lacks: (i) a *determinate yet anchorless* referent, given a closed-form construction (the load-weighted Fréchet mean on a moral manifold, the geometric generalization of Millman’s theorem); (ii) a **reflexive control dynamics** in which agents re-weight their own valuations and thereby move the reference they are measured against, recasting moral change as feedback rather than convergence toward a fixed target; and (iii) a *dissolution* rather than a solution of the is/ought problem — the far bank of Hume’s gap is shown to be a mirage of the anchoring assumption. ERT is deflationary about authority and realist about structure. It is also falsifiable, and we state the conditions that would refute it.

1. The shared assumption

Ask what makes an action good and two answers dominate. The **realist** says: there is a fact of the matter, binding independently of what anyone thinks — perhaps grounded in the nature of reasons, perhaps in God. The **anti-realist** says: there is no such fact; “good” expresses an attitude, encodes an error, or marks an arbitrary brute commitment that bottoms out a regress (one cannot rank two values without appealing to a third, *ad infinitum*).

These positions look opposed, but they agree on something deeper. Both assume that for a moral reference to be *real* — determinate, non-arbitrary, action-guiding — it must be **anchored** to something outside the system of valuers: a transcendent ground, a Platonic fact, a divine will. The realist affirms the anchor; the anti-realist, having looked and found none, infers that the reference itself is empty. The dilemma — *binding external fact, or arbitrary brute* — is generated by the anchoring assumption, not by morality.

This paper denies the assumption. We claim that a moral reference can be **determinate without being anchored**, and that the model for such a thing is already familiar to any electrical engineer.

A note on method. ERT is offered in the spirit of *Philosophy Engineering*: the discipline of stating a philosophical claim formally enough that it can be computed, made to predict, and broken. Where traditional metaethics argues toward a position and rests, ERT is built to be *operated* — it gives the moral reference a closed-form construction (the load-weighted Fréchet mean, §3), derives consequences that could have come out otherwise (the conservation law, §5.2; the schism threshold, §5.6), exposes a falsification battery (§8), and ships with a reproducible computational artifact that both verifies the formal results numerically and applies them to real moral-judgment data — continuing the program of the Geometric Ethics framework [Bond 2026]. A metaethics that cannot be wrong in public is, on this view, not yet finished; the rest of this paper is an attempt to make ERT wrong and a report of where it survives.

2. The model: a floating neutral

In a polyphase power system, several live conductors (the *phases*) drive a set of loads that meet at a common point, the *neutral*. When the neutral is *bonded* — tied to an external ground — its potential is clamped to zero; it has an anchor, and the phase voltages are measured against it.

Remove the bond. The neutral now **floats**. One might expect it to become undefined, or to wander arbitrarily. It does neither. The floating neutral sits at a perfectly definite potential, given by **Millman’s theorem** [Millman 1940]:

$$V_N = \frac{\sum_i Y_i V_i}{\sum_i Y_i}, \quad Y_i = 1/Z_i,$$

the **admittance-weighted average** of the phase voltages V_i , where Y_i is the load on phase i . No external reference appears anywhere in this expression. The neutral’s location is fixed *entirely by the loads* — by the system’s own internal configuration. It is determinate, measurable, and computable; it is also anchorless. Shift the loads and it moves; balance them and it centers; lose a load and it lurches. It is a real point with no ground beneath it.

Two facts about the floating neutral do the philosophical work.

First: it has a position but no authority. The neutral does not *command* the phases; it is simply where they net out. A phase is under no obligation to “honor” the neutral. Yet the neutral is not inert: a badly displaced neutral — under a severe load imbalance — drives overvoltages that *destroy equipment*. It has consequences without commands. Position, not authority.

Second: it is endogenous. Ask “what grounds the neutral’s location?” and the only honest answer is *nothing external grounds it; it is constituted by the loads*. Asking what *authorizes* it is a category error, like asking what authorizes the center of mass of a galaxy.

ERT is the claim that **moral reference is a floating neutral**.

3. Endogenous Reference Theory

Let a moral system comprise agents (or, more abstractly, value-bearing positions) indexed by i . Each contributes a *valuation* v_i — a position in a moral state space — held with a *weight* w_i (its intensity, prevalence, or “load”). The system’s moral reference is the endogenous point

$$R = \arg \min_x \sum_i w_i d(x, v_i)^2,$$

the **weighted Fréchet mean** [Fréchet 1948] of the valuations under the metric d of the state space. On a flat space this reduces exactly to Millman’s weighted average [Millman 1940]; on the curved **moral manifold** of Geometric Ethics [Bond 2026] (a nine-dimensional stratified space with a context-dependent metric), it is its geometric generalization. R is determinate given the loads, computable, and anchorless: no term outside the set of valuers enters its definition.

ERT then makes a sequence of claims:

1. **Realism about the reference.** R is a real feature of the system — not a projection, not an error, not a fiction. Two analysts with the same load data compute the same R . The anti-realist’s “no determinate fact” is false: there is a fact, the location of R .
2. **Deflationism about authority.** R has no normative force. It does not bind any agent to itself. The realist’s “binding fact” is false: R commands nothing.
3. **“Ought” as displacement.** Normative language reports the relation of a state to R . “ a is wrong” says, roughly, *a lies far from R along load-bearing dimensions* — it is a **strain reading**, not a decree. The intensity of moral feeling tracks displacement magnitude the way strain voltage tracks neutral shift.
4. **Endogeneity.** R is constituted by the loads. There is no meta-level — no anchor, no transcendent ground, no “moral law” above the system. *There is only the configuration, and the point it determines.*

The user’s slogan captures it exactly: *there is no bindingness; there is no meta-anything; there is only reality*. ERT is the formal content of that slogan.

4. The is/ought gap, dissolved

Hume [1739] observed that one cannot validly derive an *ought* from an *is*. The realist treats this as a gap to be bridged; the anti-realist, as a proof that oughts are not facts. ERT treats it as a **mislocated demand**.

The gap is the distance between descriptive fact and a *binding* prescription — a prescription with authority, transmitted from somewhere. But on ERT there is no such bank on the far side to reach. The floating neutral has position, not authority; “ought” is the *strain*, which is itself an is-fact (a displacement is a measurable quantity). There is no leap from is to ought because “ought,” properly understood, was an is all along — the is of *displacement from the endogenous reference*.

This is a *dissolution*, and it has a cost we state plainly (§7): one who uses “ought” to mean *categorically binding* will say we have changed the subject. We have — deliberately. The categorical bind was the anchoring assumption’s shadow; remove the anchor and the shadow goes with it. What remains — real reference, real displacement, real consequences — is everything morality needs to *function*, and nothing it needs to *command*.

5. Reflexive dynamics: a formalization

Here ERT departs decisively from a static picture, and from its electrical metaphor. In a power system a phase cannot choose its own impedance. **Agents can.** A valuer can re-weight its own valuation — come to care differently — and thereby move the reference R against which it, and everyone else, is measured. The system *steers itself relative to its own floating point*. We make this precise.

5.1 State and reference

Let agents be indexed $i = 1, \dots, N$, each with a valuation $v_i \in \mathbb{R}^n$ (a local chart of the moral manifold M) and a non-negative weight (load) w_i , with total $W = \sum_i w_i$. The endogenous reference is the weighted Fréchet mean [Fréchet 1948],

$$R = \arg \min_x \sum_i w_i d(x, v_i)^2,$$

which in the Euclidean case reduces exactly to the Millman point [Millman 1940],

$$R = \frac{1}{W} \sum_i w_i v_i = \langle v \rangle_w.$$

We work in the flat case for the analytic results and return to the manifold in §5.5.

5.2 The law of reference motion

Let valuations and weights evolve smoothly in time. Differentiating $R = (\sum_i w_i v_i)/W$ gives the central identity

$$\boxed{\dot{R} = \underbrace{\langle \dot{v} \rangle_w}_{\text{position drift}} + \underbrace{\frac{1}{W} \sum_i \dot{w}_i (v_i - R)}_{\text{reweighting}}} \quad (\star)$$

where $\langle \dot{v} \rangle_w = \frac{1}{W} \sum_i w_i \dot{v}_i$. The reference moves by exactly two routes: a net weighted **drift** of valuations, or a **reweighting** that loads one side of the current reference (a covariance between weight change and displacement). Nothing external can move it — consistent with endogeneity.

Proposition 1 (Conservation of the reference under homogeneous conformity). *If weights are fixed ($\dot{w}_i = 0$) and every agent relaxes toward the reference at a common rate, $\dot{v}_i = k(R - v_i)$ with $k > 0$, then $\dot{R} = 0$. The reference is invariant, and the configuration converges to consensus at $R(0)$, with dispersion $\sum_i w_i \|v_i - R\|^2$ decaying as e^{-2kt} .*

Proof. By $(\star)$, $\dot{R} = \langle \dot{v} \rangle_w = \frac{1}{W} \sum_i w_i k(R - v_i) = k(R - \langle v \rangle_w) = k(R - R) = 0$. With R fixed, $\frac{d}{dt}(v_i - R) = -k(v_i - R)$, so each displacement decays as e^{-kt} and the weighted dispersion as e^{-2kt} . $\square$

This has a sharp consequence:

Corollary (Progress is not free). *A purely conformist population cannot move its own moral reference.* Reference motion requires an **asymmetry**: heterogeneous responsiveness, weight dynamics, nonlinearity, or exogenous forcing. The floating neutral does not drift on its own; something must load it.

The heterogeneous case makes the mechanism explicit. With $\dot{v}_i = k_i(R - v_i)$ and fixed weights,

$$\dot{R} = \frac{\sum_i w_i k_i}{W} (R - R_k), \quad R_k = \frac{\sum_i w_i k_i v_i}{\sum_i w_i k_i},$$

so the reference drifts **toward the responsiveness-weighted mean** R_k — i.e., toward the positions of the agents who update fastest and are weighted most heavily. The neutral is pulled by whoever moves first and loudest, not by who is “right.”

5.3 Regimes

Add a *conviction* (differentiation) term that pushes an agent away from the reference — identity, contrarianism, principled resistance:

$$\dot{v}_i = \underbrace{k_i(R - v_i)}_{\text{conformity}} - \underbrace{c_i \phi(v_i - R)}_{\text{conviction}}, \quad \dot{w}_i = \alpha \psi(v_i - R).$$

The qualitative outcomes mirror those of bounded-confidence opinion dynamics [Hegselmann–Krause 2002] — consensus, polarization, or fragmentation — selected by the conformity/conviction balance and the weight-feedback gain α :

- **Conformity-dominated** ($c_i \approx 0$): consensus at a conserved neutral (Prop. 1).
- **Balanced**: a stable *dispersed* equilibrium — persistent disagreement around a stable reference (a “moral ecology”), not collapse to unanimity.
- **Conviction- or feedback-dominated**: polarization (bimodal split) or **limit cycles** (moral oscillation), as the reweighting term in $(\star)$ drives and is driven by displacement.

Shocks. If a faction’s load vanishes abruptly, $w_j \rightarrow 0$, the reference lurches by $\Delta R \approx -\frac{w_j}{W}(v_j - R)$ — the neutral jumps away from the departed position, and the remaining agents experience a sudden jump in their own displacement: the moral analogue of losing a phase and driving **overvoltage** on the rest. This models norm collapse, generational turnover, and revolution as the *same* mechanism, differing only in rate.

5.4 Equilibria and stability

A self-consistent equilibrium is a configuration $\{v_i^*, w_i^*\}$ with $\dot{v}_i = \dot{w}_i = 0$ and $R = R(\{v_i^*, w_i^*\})$. The system is **self-referential**: each agent’s motion feeds back through the mean, $\partial R / \partial v_j = w_j / W$, so the Jacobian is not block-diagonal. The consensus equilibrium ($v_i = R$, conformity only) has eigenvalues $-k_i$ and is asymptotically stable. Dispersed equilibria are stable when conformity dominates conviction at the fixed point; crossing that balance produces a Hopf bifurcation into the oscillatory regime. We state these as the structure of the analysis; the full classification on M is open (§5.8).

5.5 Relation to existing dynamics, and what is actually new

Honesty requires naming the neighbors. Equation $(\star)$ and the regimes of §5.3 are close kin to consensus and opinion-dynamics models — DeGroot averaging [DeGroot 1974], the Friedkin–Johnsen extension, and bounded-confidence dynamics [Hegselmann–Krause 2002]. **ERT does not claim to invent this mathematics.** Its contributions over that literature are three, and they are conceptual before they are technical:

1. **Interpretation.** The consensus point is reinterpreted as the *moral reference* — the floating neutral — and “ought” as displacement from it. The opinion-dynamics literature studies *where opinions go*; ERT reads the same object as *the metaethical reference itself*, with no anchor and no authority.
2. **Loads as first-class.** Standard models fix influence weights or set them exogenously; ERT makes the weight dynamics $\dot{w}_i$ co-equal with $\dot{v}_i$, which is what equation $(\star)$ shows is *necessary* for the reference to move at all (the Corollary). Reflexive reweighting is the engine of moral change.
3. **Curvature.** Lifting to the moral manifold M , R is the weighted Fréchet/Karcher mean [Karcher 1977], whose existence and uniqueness require curvature and locality conditions [Afsari 2011]. Disagreement spread across a region of high curvature or near a cut locus can make R non-unique — a genuinely new phenomenon with no flat-space or scalar analogue: *the reference can bifurcate*, a candidate formal model of moral schism.

The one result we do claim as ERT’s own is the **conservation law $(\star)$ together with Proposition 1 and its Corollary**, read metaethically: a self-referencing valuing system cannot move its own reference without an internal asymmetry. That is a falsifiable, non-obvious statement about moral change, and it falls directly out of taking the floating-neutral picture seriously. The second is the schism result below, which has no scalar or flat-space analogue at all.

5.6 Schism: bifurcation of the moral reference

The reference R minimizes the Fréchet function $F(x) = \frac{1}{2} \sum_i w_i d(x, v_i)^2$. Whether R is *unique* is not a metaethical stipulation — it is a fact about the **curvature** of the value space, and it is where ERT makes a prediction no flat or scalar theory can.

The geometry. For a single datum p , the Hessian of $\frac{1}{2}d(\cdot, p)^2$ on a space of constant sectional curvature κ has eigenvalue 1 along the geodesic toward p and, in the orthogonal directions,

$$h_\kappa(d) = \sqrt{\kappa} d \cot(\sqrt{\kappa} d) \quad (\kappa > 0), \quad h_0(d) = 1, \quad h_\kappa(d) = \sqrt{-\kappa} d \coth(\sqrt{-\kappa} d) \quad (\kappa < 0).$$

The Fréchet Hessian is $H(x) = \sum_i w_i \text{Hess } \frac{1}{2}d(\cdot, v_i)^2$. For $\kappa \leq 0$, $h_\kappa \geq 1$, so F is strictly geodesically convex and R is **globally unique** (the Hadamard case). For $\kappa > 0$, $h_\kappa(d)$ falls from 1, reaches 0 at $d = \pi/(2\sqrt{\kappa})$, and goes negative beyond — so F can lose convexity and acquire multiple minima once valuations spread. **Schism is therefore a strictly positive-curvature phenomenon: on flat or negatively curved value space, geometry alone cannot split the reference.**

Worked case (exactly solvable). Put equal loads at A, B on the space form S_κ^n , at geodesic separation 2δ . By symmetry the midpoint m is a critical point of F ; its Hessian is 1 along the A – B axis and $h_\kappa(\delta) = \sqrt{\kappa} \delta \cot(\sqrt{\kappa} \delta)$ orthogonal to it. Hence:

- $\sqrt{\kappa} \delta < \pi/2$ — the midpoint is the **unique** reference. *Ordinary pluralism*: people disagree, but share one neutral.
- $\sqrt{\kappa} \delta \rightarrow \pi/2$ — the orthogonal stiffness $\rightarrow 0$; the reference goes **degenerate** (a whole subsphere of equally good neutrals): the bifurcation point.
- Beyond it (resolved by a third cluster, asymmetric weights, or variable curvature) the degeneracy breaks into **discrete competing references** — a saddle-node/pitchfork in which the single center becomes a saddle and sub-community neutrals split off around it.

Proposition 2 (Schism threshold). *Let the loads lie in a region of M with sectional curvature $\leq \kappa$ and (weighted) dispersion radius ρ . If $\rho < \rho^* = \frac{1}{2} \min\{\text{inj}(M), \pi/\sqrt{\kappa}\}$, the moral reference is unique [Karcher 1977; Afsari 2011]. If $\rho > \rho^*$, the Fréchet function may admit multiple minimizers — the reference bifurcates. For $\kappa \leq 0$, $\rho^* = \infty$: no geometric schism is possible.*

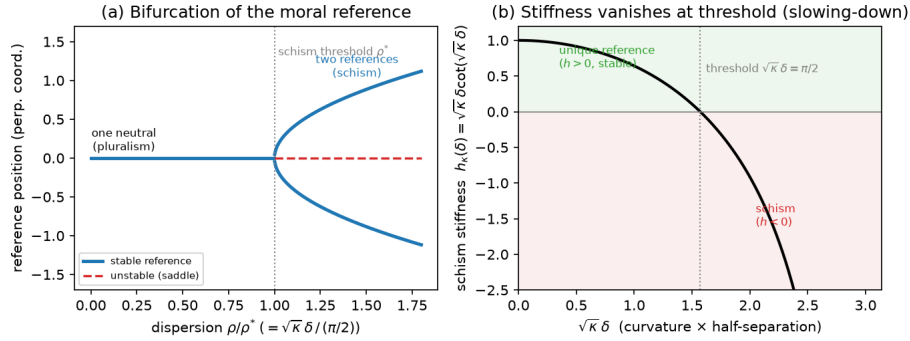


Figure 1: **Schism as a bifurcation of the moral reference.** (a) As dispersion ρ crosses the curvature-set threshold ρ^* , the single floating neutral loses stability (dashed) and two competing references split off — a pitchfork. (b) The exact schism stiffness $h_\kappa(\delta) = \sqrt{\kappa} \delta \cot(\sqrt{\kappa} \delta)$ for the symmetric two-cluster sphere falls to zero at $\sqrt{\kappa} \delta = \pi/2$; its vanishing is the critical slowing-down of §5.7.

Why this is the right picture of schism. Positive curvature in value space is exactly the regime where the geodesic *compromise* between two positions is **worse than either pole** — where splitting the difference satisfies no one. That is the signature of sacred or protected values, on which half-measures read as betrayal. Negative or flat curvature is the regime of ordinary, tradeable values, where intermediate positions are stable. ERT therefore predicts that **schisms concentrate on the high-curvature (sacred/identity) axes of the moral manifold and not on the tradeable ones** — and it explains schism *without* either faction being in error (contra error theory) and *without* an external arbiter

(no anchor): the two post-schism communities are simply different stable basins of the same Fréchet landscape, each internally coherent. The split is geometric, not moral.

5.7 Critical slowing-down: an early-warning signal

The threshold is not just a static fact; coupled to the reflexive dynamics of §5.2–5.4 it becomes *predictive*. As a population’s dispersion drifts toward ρ^* (driven by the reweighting term in $(\star)$), the orthogonal stiffness $h_\kappa(\delta) \rightarrow 0$: the reference’s restoring curvature vanishes. This is the generic precursor of a saddle-node/pitchfork bifurcation — **critical slowing-down** — and it yields measurable early-warning signals *before* a schism actually occurs:

1. **Rising recovery time** — the aggregate reference relaxes ever more slowly after shocks (autocorrelation of R rises toward 1);
2. **Rising variance/volatility** of R under constant perturbation;
3. **Flattening** of the Fréchet landscape around R (the smallest Hessian eigenvalue trends to zero).

Operationally, on longitudinal moral-survey or corpus time-series, a community approaching schism should show its reference becoming *more volatile and slower to recover* in the run-up to the split — a prediction sharply distinct from “the groups simply drifted apart,” and testable with the same instruments as §6. Schism, on this account, is forecastable.

Honest caveats. The closed-form thresholds assume constant curvature and the symmetric configuration; the real moral manifold is stratified with variable curvature, so ρ^* is *local* and cut-locus structure matters (§5.8). The curvature $\leftrightarrow$ sacred-value correspondence is a conjecture to be measured, not a result. And “multiple Karcher means exist” is necessary, not sufficient, for an observed social schism — the dynamics must also carry sub-populations into the distinct basins.

5.8 Open problems

Existence, uniqueness, and stability of the self-referential equilibrium on M ; the full bifurcation classification under joint (v, w) dynamics; conditions for Fréchet-mean uniqueness on the *stratified* moral manifold (cut-locus and curvature bounds beyond [Afsari 2011]); and the empirical estimation of k_i, c_i, w_i and the recovery of $(\star)$ from longitudinal moral-survey data. The last is the bridge back to the corpus evidence of §6 and to the falsification tests of §8.

6. Evidence: why references cluster

If R is endogenous, distinct societies are distinct load configurations and should, in general, determine distinct references. Why, then, the striking cross-cultural agreement on core norms?

ERT’s answer is mechanical, not metaphysical: **human load configurations**

cluster, so their neutrals cluster. Shared embodiment, shared social pressures, and shared dependence on cooperation make the $\{v_i, w_i\}$ of independent cultures similar enough that their weighted Fréchet means fall near one another — *without* any common external anchor pulling them there. Convergence is evidence of *similar loads*, not of a shared ground.

This reframes the empirical program of Geometric Ethics. Analysis of a large cross-lingual normative corpus (109,000 passages across eleven languages and roughly three millennia) [Bond 2026] finds that the *dimensional structure* of moral evaluation is stable across languages and eras — the same low-dimensional axes recur regardless of framing. On ERT, that stability is the **phase structure** (the dimensions along which loads can sit), and the cross-cultural agreement on particular verdicts is the **clustering of neutrals** within it. The program *measures the neutral*; it does not legislate it.

One honest caveat carries over: the strongest decoding results are measured *through* a multilingual embedding model whose language-agnostic representation may itself carry some of the apparent universality. Until that confound is excluded, “the axes are invariant” retains a deflationary reading — which ERT can absorb (the instrument would then be measuring a real but partly representational structure) but should not hide.

7. Relation to existing positions

ERT is not error theory: Mackie [1977] holds there are *no* moral facts, only systematic error; ERT holds there is a determinate fact — the location of R — that simply lacks authority. It is not quasi-realism (Blackburn [1998]): R is computed from real loads, not projected from attitudes and then talked about *as if* real. It is closest in spirit to **constructivism** (Street [2006]; Korsgaard [1996]) — morality as constituted by valuers rather than found — but supplies what constructivism characteristically lacks: an *explicit, falsifiable construction* (the weighted Fréchet mean) and a *dynamics* (the control law). It is compatible with naturalist programs (Railton [1986]; Boyd [1988]) and with “morality as cooperation” (Curry et al. [2019]), which it can read as accounts of *why the loads sit where they do*. Its distinctive commitments are the two halves held together: **realism about the reference, deflationism about authority** — a combination the standard taxonomy does not have a box for, because the taxonomy presupposes the anchoring assumption ERT denies.

8. Falsifiability

In the Philosophy Engineering tradition, a metaethical claim earns its keep by entailing testable predictions. ERT would be undermined if:

1. **Reference indeterminacy.** Moral systems with stable, shared load configurations nonetheless exhibited *no* determinate reference — verdicts scattering with no computable center. (ERT predicts a determinate R where-

ever loads are well-defined.)

2. **Load-independence.** The reference failed to track the loads — e.g., shifting valuations across a population left aggregate moral verdicts unchanged, or vice versa. (ERT predicts R moves with $\{v_i, w_i\}$ as Millman/Fréchet prescribes.)
3. **Anchor signature.** Cross-cultural convergence persisted *even where loads diverge* — references coinciding despite dissimilar configurations — which would be evidence of an external attractor (an anchor) after all, contra endogeneity.
4. **Dynamical mismatch.** Observed moral change failed to obey a feedback law of the stated form — e.g., displacement from the current reference had no predictive relation to subsequent valuation shifts.
5. **Schism without curvature.** Communities fractured into stable, competing references *independently of value-space curvature* — splitting as readily on low-curvature (tradeable) axes as on high-curvature (sacred/identity) ones, or splitting while dispersion stayed well below the estimated threshold ρ^* . (ERT predicts schism is confined to high-curvature axes and to $\rho > \rho^*$; §5.6.) *Concretely:* estimate the metric and curvature from the corpus/survey embedding, locate the high- κ axes, and test whether documented schisms (denominational splits, party realignments, professional-ethics fractures) concentrate there and cross the threshold — not on the low- κ ones.
6. **Schism without warning.** Observed splits arrived as abrupt jumps with *no* precursor — no rise in the reference’s recovery time or variance beforehand — or those critical-slowness signals appeared just as often in communities that did **not** split. (ERT predicts a measurable slowing-down as $\rho \rightarrow \rho^*$; §5.7.) *Concretely:* on a moral time series, compute the lag-1 autocorrelation and variance of the aggregate reference in a sliding window; ERT requires both to trend upward in the run-up to a schism and to be absent in matched non-splitting controls.

These are demanding and, with population-scale moral measurement, approachable. Tests 5–6 are the sharpest: they target the schism result (§5.6–5.7), which has no scalar or flat-space analogue, so a clean negative there would falsify ERT’s distinctive content specifically — not merely some shared feature of structural metaethics.

Preliminary results. The accompanying artifact exercises these tests rather than leaving them as promissory notes. The synthetic confirmation is clean: the schism stiffness $h_\kappa(\delta)$ matches its closed form to four decimals and vanishes at $\sqrt{\kappa}\delta = \pi/2$, and the Fréchet reference bifurcates only under positive curvature — Proposition 2 holds numerically. A first probe of Test 5 on real moral judgments (2,900 labelled AITA dilemmas) was, by contrast, **inconclusive**: contestation showed no relation to bimodality of the recovered moral-position axis (robust across two encoders), but the test is underpowered — the moral axis was only weakly recoverable (AUC = 0.60) on a heavily verdict-skewed corpus, and the curvature half of Test 5 was not exercised. We report this as a null

the instrument cannot yet turn into a falsification; a decisive test needs a polarized, vote-margin corpus and a direct curvature estimate. In the Philosophy Engineering spirit, the failure is part of the record, not hidden.

Update — the sacred-axis prediction is falsified. The decisive Test-5b run on Moral Machine AMCE preferences (177 agents: 176 LLMs plus one aggregate human, labelled 9-D — no embedding, no AUC ceiling) found that agent references *do* cluster into ≥ 2 basins (GMM BIC-optimal at 4 components; silhouette 0.36–0.40), **but the multimodality concentrates on the impartial/consequentialist axes** (number of lives, species), not the **sacred/identity** ones (age, gender, status, fitness), which show consensus — Mann–Whitney sacred $>$ impartial $p = 0.90$ (the wrong direction). Combined with the GlobalOpinionQA null ($n = 26$), ERT’s distinctive prediction — that schism concentrates on sacred/high-curvature axes (§5.6) — has **zero confirmations and is falsified on the best-powered instrument available**. What survives is only the weaker claim that moral references bifurcate into camps, which is ordinary opinion-dynamics clustering and not unique to ERT. (*Caveat: the exact per-country human AMCE matrix was unavailable — only an aggregate human column — so the human-cultural version is technically untested; but the mechanism failed on a strong, well-powered proxy population on top of the GOQA null.*) ERT therefore rests on its philosophical reframe (§§1–4) and its theorems (§5.2, §5.6), **not** on this empirical signature; the schism result should be read as a falsifiable conjecture that did not survive, reported in full per the Philosophy Engineering commitment in §1.

9. Limits and open problems

We close with what ERT does *not* do. It **deflates** bindingness; those who want morality to *command* will find that demand unmet — by design, and they may reasonably regard this as changing the subject rather than answering it. The **reflexive dynamics** of §5 are stated, not solved; the existence, stability, and uniqueness conditions of the self-referential equilibrium are open. The **measurement** of R inherits the embedding-instrument confound of §6. And there is a genuine **normativity-for-the-individual** residue: ERT explains where the system’s reference *is* and why one *feels* strain, but an agent can still ask “why should *I* reduce my displacement?” — to which ERT’s only answer is the honest one: *no external authority requires you to; the consequences of large displacement are simply real* — the overvoltage does not ask permission before it burns.

That answer will not satisfy everyone. But it is the answer the floating neutral actually gives, and ERT’s contribution is to give it precisely: a moral reference that is real, computable, endogenous, consequential — and anchorless.

References

- Afsari, B. (2011). Riemannian L^p center of mass: existence, uniqueness, and convexity. *Proceedings of the American Mathematical Society*, 139(2), 655–673.
- Blackburn, S. (1998). *Ruling Passions: A Theory of Practical Reasoning*. Oxford University Press.
- Bond, A. H. (2026). *Geometric Ethics: A Foundational Theory of Moral Structure* (and the accompanying cross-lingual corpus analysis, 109,294 passages, 11 languages). Working paper / ErisML foundations.
- Boyd, R. (1988). How to Be a Moral Realist. In G. Sayre-McCord (Ed.), *Essays on Moral Realism* (pp. 181–228). Cornell University Press.
- Curry, O. S., Mullins, D. A., & Whitehouse, H. (2019). Is It Good to Cooperate? Testing the Theory of Morality-as-Cooperation in 60 Societies. *Current Anthropology*, 60(1), 47–69.
- DeGroot, M. H. (1974). Reaching a Consensus. *Journal of the American Statistical Association*, 69(345), 118–121.
- Fréchet, M. (1948). Les éléments aléatoires de nature quelconque dans un espace distancié. *Annales de l’Institut Henri Poincaré*, 10(4), 215–310.
- Hegselmann, R., & Krause, U. (2002). Opinion Dynamics and Bounded Confidence: Models, Analysis, and Simulation. *Journal of Artificial Societies and Social Simulation*, 5(3).
- Hume, D. (1739–40). *A Treatise of Human Nature*. (Is/ought: Book III, Part I, §I.)
- Karcher, H. (1977). Riemannian center of mass and mollifier smoothing. *Communications on Pure and Applied Mathematics*, 30(5), 509–541.
- Korsgaard, C. M. (1996). *The Sources of Normativity*. Cambridge University Press.
- Mackie, J. L. (1977). *Ethics: Inventing Right and Wrong*. Penguin.
- Millman, J. (1940). A Useful Network Theorem. *Proceedings of the IRE*, 28(9), 413–417.
- Railton, P. (1986). Moral Realism. *The Philosophical Review*, 95(2), 163–207.
- Street, S. (2006). A Darwinian Dilemma for Realist Theories of Value. *Philosophical Studies*, 127(1), 109–166.

Next steps: classify the self-referential equilibria on M (§5.4, §5.8); design the load-independence and anchor-signature tests (§8); add a schism-threshold / early-warning test to §8; reconcile the reference R with the Geometric Ethics manifold formalism [Bond 2026]; and attempt empirical recovery of $(\star)$ from longitudinal moral-survey data.